



Retail Inventory **Risk & Performance** **Intelligence** **Dashboard**

(Power BI)

Project Overview

This project presents an **end-to-end Retail Business Intelligence Dashboard** built using Power BI to analyze **sales performance, inventory health, supplier reliability, and store profitability**. The objective was to design a **structured relational data model**, implement **robust DAX measures**, and build **executive-level analytics** that connect sales performance with inventory flow and supply chain reliability.

The dashboard integrates **sales analytics, inventory risk monitoring, supplier supply analysis, and profitability insights** to simulate a **real-world retail business intelligence environment**. It provides a **360° view of retail operations**, enabling decision-makers to track revenue drivers, detect inventory risks, and evaluate supply chain efficiency.

Business Objectives

- ★ Measure overall retail sales and profitability performance

- ★ Monitor inventory health and detect stockout risks
- ★ Identify high-performing stores and product categories
- ★ Analyze supplier contribution to inventory supply
- ★ Track supply vs demand balance across the business
- ★ Detect inventory risk patterns before stockouts occur
- ★ Provide actionable insights for retail operations and supply chain planning

Dataset Description

The dataset consists of structured retail system tables representing **sales transactions, inventory management, product catalog, store information, and supplier operations.**

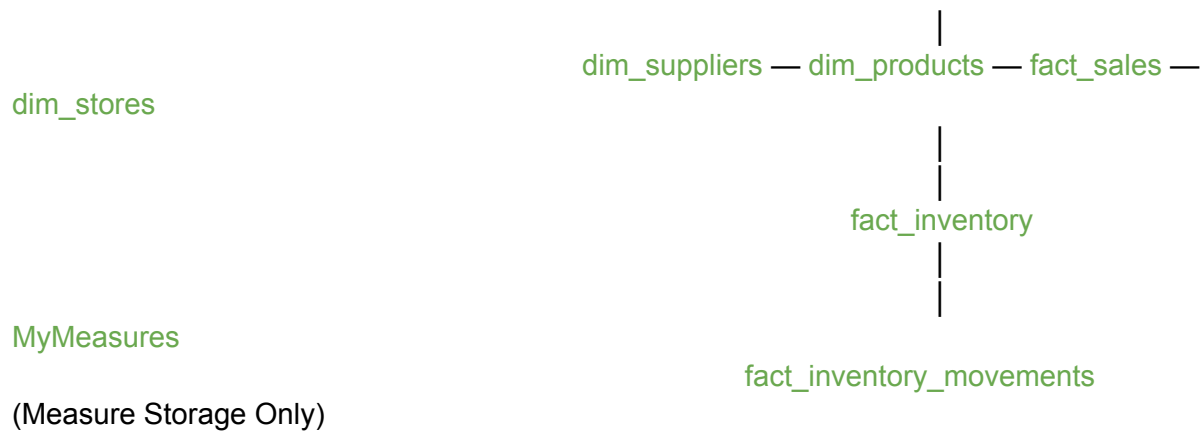
Table	Description
fact_sales	Sales transactions including revenue, cost, and units sold
fact_inventory	Inventory availability and stock levels
fact_inventory_movements	Inventory inflow and outflow movements
dim_products	Product catalog, categories, and supplier linkage
dim_stores	Store location, region, and store attributes
dim_suppliers	Supplier details and supply chain relationships
dim_date	Calendar table used for time intelligence
MyMeasures	Dedicated table to store all DAX measures

The dataset simulates a **real-world retail data warehouse environment.**

Data Model Design

The model follows a **clean star-schema structure** for optimal performance and analytical flexibility.

dim_date
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Model Characteristics

- ❖ One-to-Many relationships maintained
- ❖ Single-direction filtering applied across fact tables
- ❖ Dimension tables provide consistent filtering context
- ❖ No ambiguous relationships or circular dependencies
- ❖ Dedicated **MyMeasures table** used for centralized DAX management

The model follows **retail analytical architecture**:

- **fact_sales** acts as the primary sales performance table
- **fact_inventory** tracks inventory stock levels
- **fact_inventory_movements** tracks inventory inflow and outflow
- **dim_products** connects products with categories and suppliers
- **dim_stores** provides store-level segmentation
- **dim_suppliers** enables supplier contribution analysis



Key DAX Measures Implemented

1 Core Business KPIs

- ❖ Total Revenue
- ❖ Total Profit
- ❖ Profit Margin (%)
- ❖ Total Units Sold
- ❖ Sales Last Year
- ❖ Year-on-Year Growth (%)

2 Inventory & Risk Analysis

- ❖ Inventory Health Score
- ❖ SKU Critical Risk Count
- ❖ Inventory vs Reorder Point
- ❖ Inventory Risk by Category
- ❖ Stockout Risk Indicator

3 Supply Chain & Inventory Flow

- ❖ Stock In
- ❖ Stock Out
- ❖ Net Inventory Flow
- ❖ Supplier Inventory Contribution
- ❖ Supplier Stockout Risk Exposure

4 Profitability & Store Performance

- ❖ Store Level Profit
- ❖ Category Profit Contribution
- ❖ High Margin Product Categories
- ❖ Profit Distribution by Region

All measures were implemented using **correct filter context handling and aggregation logic** to ensure KPI integrity across slicers and filters.



Dashboard Pages

1 Executive KPI & Strategic Insights

- ❖ Total Revenue
- ❖ Total Profit
- ❖ Profit Margin (%)
- ❖ Total Units Sold
- ❖ Top Performing Store
- ❖ Sales & Profit Trend
- ❖ Sales Distribution by Region

Purpose: Provide a **high-level executive overview of retail performance**, including revenue growth, profitability trends, and regional sales distribution.

2 Sales Performance Overview

- ❖ Inventory Health KPI
- ❖ Year-on-Year Growth (%)
- ❖ Category Performance Analysis
- ❖ Sales Performance by State (Map View)
- ❖ Inventory Risk by Category

Purpose: Analyze **sales trends and product category performance** while identifying early inventory risks.

3 Inventory Risk Intelligence

- ❖ Inventory vs Reorder Point Analysis (Scatter Plot)
- ❖ Inventory Risk Trend
- ❖ Stockout Risk by Product Category
- ❖ Supplier Stockout Risk Contribution
- ❖ Top 15 Critical SKU Risk Table

Purpose: Detect **inventory shortages, high-risk SKUs, and supplier-related stockout risks** to support proactive inventory planning.

4 Supply Chain & Inventory Flow Intelligence

- ❖ Top 10 Suppliers by Inventory Contribution
- ❖ Top Product Categories by Supplier Deliveries
- ❖ Inventory Flow Trend (Supply vs Demand)
- ❖ Net Inventory Flow KPI

Purpose: Evaluate **supply chain reliability and inventory flow dynamics**, helping detect supply-demand imbalance.

5 Store & Profitability Intelligence

- ❖ High Margin Product Categories
- ❖ Top 10 Stores by Profit
- ❖ Profit Contribution by Category
- ❖ Profit Distribution by State

Purpose: Identify *profit drivers across stores and product categories* to support better merchandising and pricing strategy.



Overall Dashboard Objective

To provide a **complete retail intelligence framework** by integrating:

- ★ Sales performance analytics
- ★ Inventory risk monitoring
- ★ Supply chain contribution analysis
- ★ Store profitability insights

This enables decision-makers to **balance revenue growth with operational efficiency and supply chain stability**.



Advanced UX Features

- ★ Collapsible slicer panel for clean dashboard layout Interactive filtering across multiple pages
- ★ Donut-based KPI progress visuals
- ★ Dynamic map-based regional sales analysis
- ★ Dedicated **MyMeasures table for DAX management**
- ★ Consistent color theme for performance indicators

The dashboard balances **analytical depth with professional visual design**.



Key Business Insights

- ★ Total Revenue and Total Profit provide a clear view of business performance.
- ★ Profit Margin remains stable across most categories, indicating a healthy pricing strategy.
- ★ Certain product categories show **higher inventory risk compared to others**.
- ★ Supplier contribution analysis reveals **inventory dependency patterns**.
- ★ Sales performance varies significantly across regions and stores.
- ★ Net Inventory Flow indicates whether supply is **outpacing demand or vice versa**.

Note: Dataset is **synthetic**, therefore some distribution patterns may appear smoother than real-world data.



Business Recommendations

- ★ Monitor categories with high inventory risk and adjust reorder strategies.
 - ★ Improve supplier diversification to reduce stockout dependency.
 - ★ Focus on high-margin categories to maximize profitability.
 - ★ Evaluate underperforming stores for operational improvements.
 - ★ Align supply chain deliveries with demand trends to avoid overstock or shortages.
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Tools Used

- ★ Power BI
 - ★ Data Modeling (Star Schema Design)
 - ★ Advanced DAX Measures
 - ★ Power Query (Data Transformation)
 - ★ Supply Chain & Retail Analytics Techniques
 - ★ Business Intelligence Visualization Design
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Portfolio Value

This project demonstrates:

- ★ End-to-end BI dashboard development
- ★ Retail analytics and inventory intelligence
- ★ Advanced DAX implementation
- ★ Supply chain analytics framework
- ★ Data modeling best practices
- ★ Business storytelling through dashboards

Suitable for roles in:

- ❖ Data Analyst
- ❖ Business Intelligence Analyst

- ❖ Retail Analytics
- ❖ Supply Chain Analytics

It reflects the ability to **transform operational retail data into strategic business insights**.



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